

Lecture 1.1 – Intro to the Course

Specific Learning Objectives:

None

Agenda for today:

1. A little about me
2. Go over syllabus
3. Why learn R?
4. Learn to train a dog
5. Install R and RStudio

Student Office Hours

Instructor: Lindsay Waldrop, Ph.D.

Grader: Eli Karmon

**Wed 1 am - 2:30 pm
Thur 11:30 am – 1 pm
by appointment**

What are office hours and why should you come?

- Office hours are for students! You are not bothering or interrupting us, this is time we set aside every week for my students!
- They are a great place to get clarification on content, ask questions, listen to other students ask questions, practice work, etc.
- They are also a great space for mentorship: career advice, info on graduate schools, exploring things that interest you, connecting with research, etc!

Course Navigation

- Where do I find course information?

Github repository: <https://github.com/CPSC-292/Fall2025-CourseInfo>

- syllabus
- course learning objectives
- lecture notes
- course schedule
- course assignments
- sample code

- Where do I find and turn in assignments and see my grades?

Course Canvas sites:

MWF 9 - 10 (Section 01): <https://canvas.chapman.edu/courses/77443>

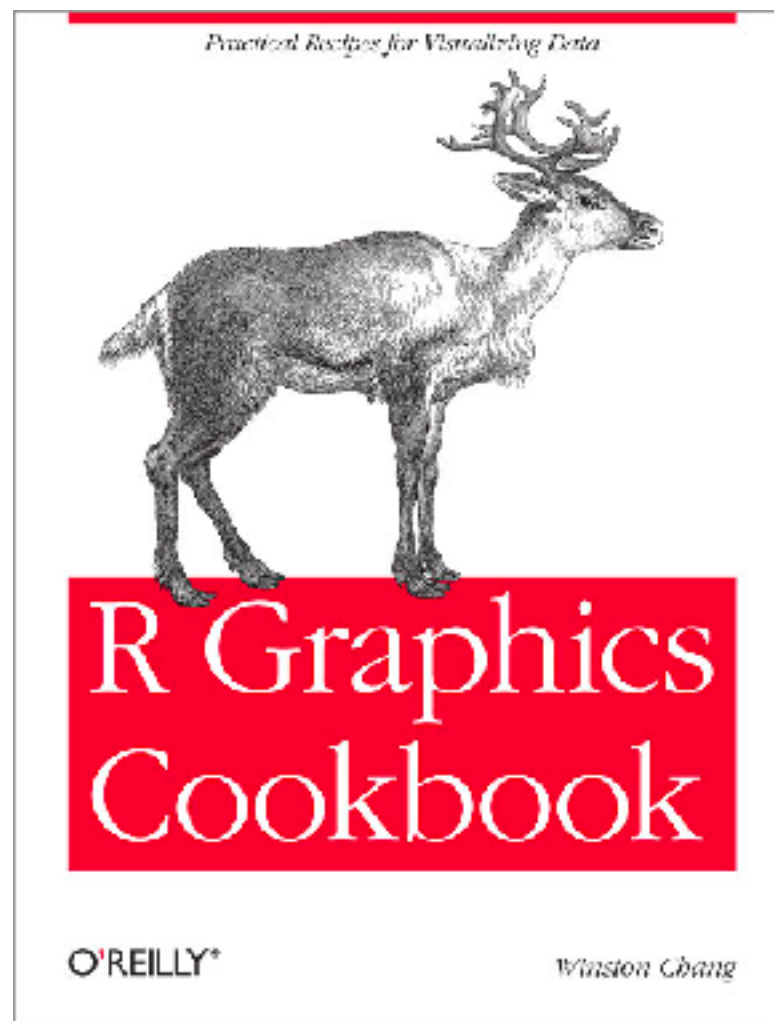
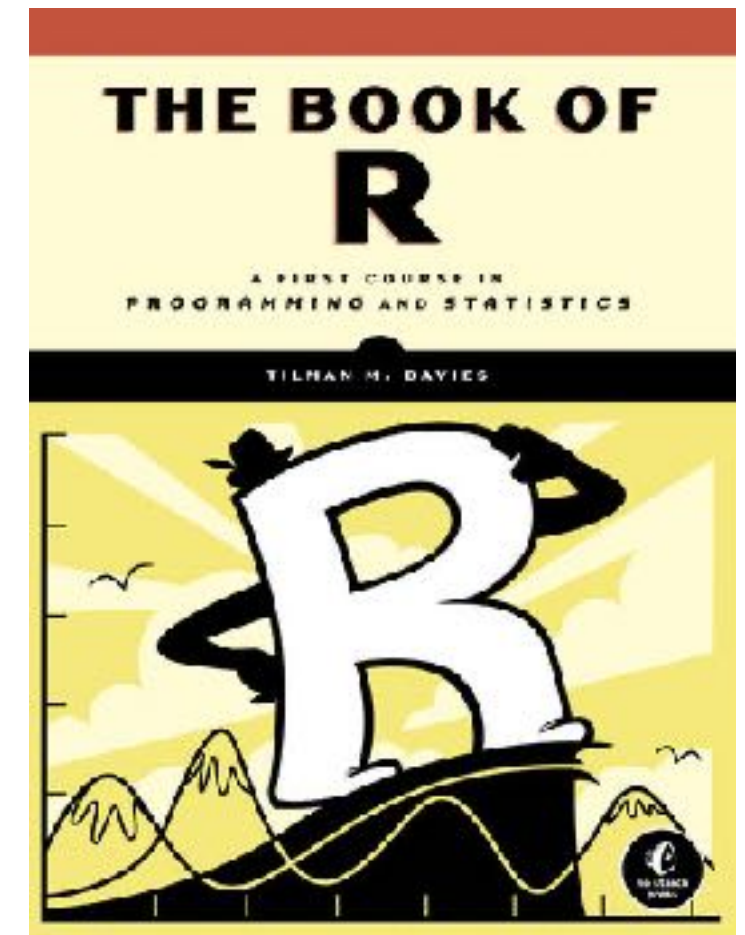
MWF 10 - 11 (Section 02): <https://canvas.chapman.edu/courses/77445>

- What is the best way to communicate?

Slack Channel! Link in Canvas Module!

Course Materials

- **Required Text:** *The Book of R* by Tilman Davies.
First Edition, No Starch Press. ISBN-13: 978-1-59327-651-5.
Link to publisher website: <https://nostarch.com/bookofr>



- **Suggested Text:** *R Graphics Cookbook* by Winston Chang.
First Edition, O'Reilly Media. ISBN 9781491978603.
Online at: <https://r-graphics.org/>

How this course works

- This is a **labor-based** and **mastery-based** course.
- Your grade in this course will be determined by:
 1. Your labor and participation.
 2. Completion of work will indicate level of mastery of course material.

Grading System (Labor)

- All work is assessed on a three-tiered scale:
 - **Completed and Satisfactory** (score of 1)
 - **Completed and Unsatisfactory** (score of 0) - you will receive feedback and have another opportunity to achieve satisfactory level.
 - **Not Completed** (score of 0) - no late work will be accepted.
- **Life Happens Clause:** request a new deadline for any assignment within 48 hours of the deadline. Request via Slack to ***your*** instructor. ***Be sure to include the new deadline (date and time) you are requesting.***

Grading System (Labor)

- **Rescoring work:** You are allowed 4 attempts for each work item (except Projects).
 - Each attempt must be made within 3 business days of receiving feedback from the instructor.
 - **It is critical that you keep up-to-date with material** (especially in Unit 1).
- **Assignments are complete only after lecture videos are viewed!**
- **Participation is required at all levels.** Non-participation is marked only. Reasons: absence, excessive lateness (>10 mins), off task, distraction, etc.

Grading System (Mastery)

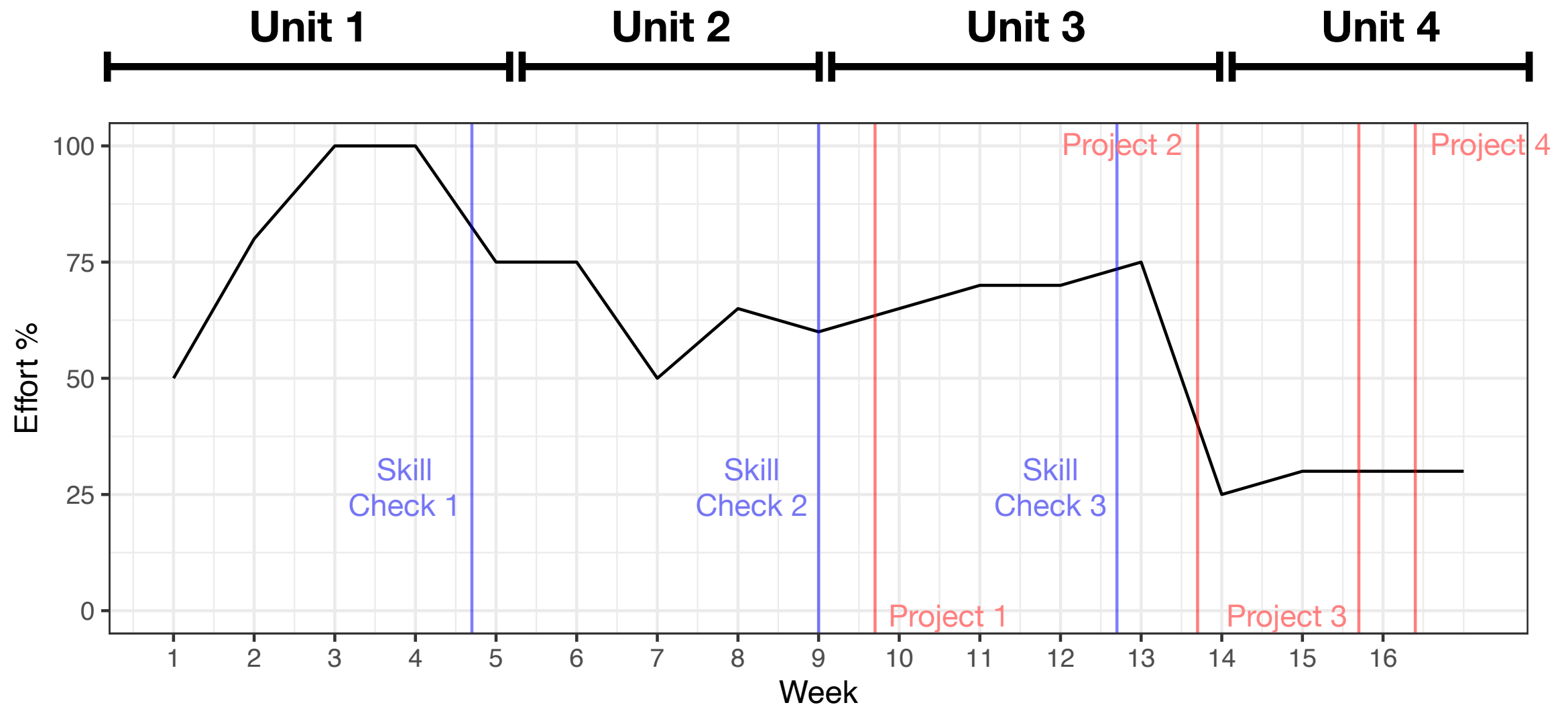
- Work is designed to demonstrate levels of mastery in the material.
- **Assignments (C-level):** Designed to assess *basic competence* of learning objectives covered. 30 in total.
- **Projects (B-level):** Designed to assess *mastery* of learning objectives covered. Work will involve synthesis, creativity, independence, and originality. 4 in total.
- **Skill Checks (A-level):** Designed to assess *advanced competence* of learning objectives covered. Usually involve synthesis of concepts and more independence than assignments. 3 in total.

Grading System (Mastery)

Grade-score guarantees table (in syllabus)

Final Course Grade	Skill Checks completed (A-level)	Projects completed (B-level)	Assignments completed (C-level)	Course Points
A	3	4	30	47
B	1	4	30	39
C	0	2	30	32
D or F	0	< 2	< 28	< 30

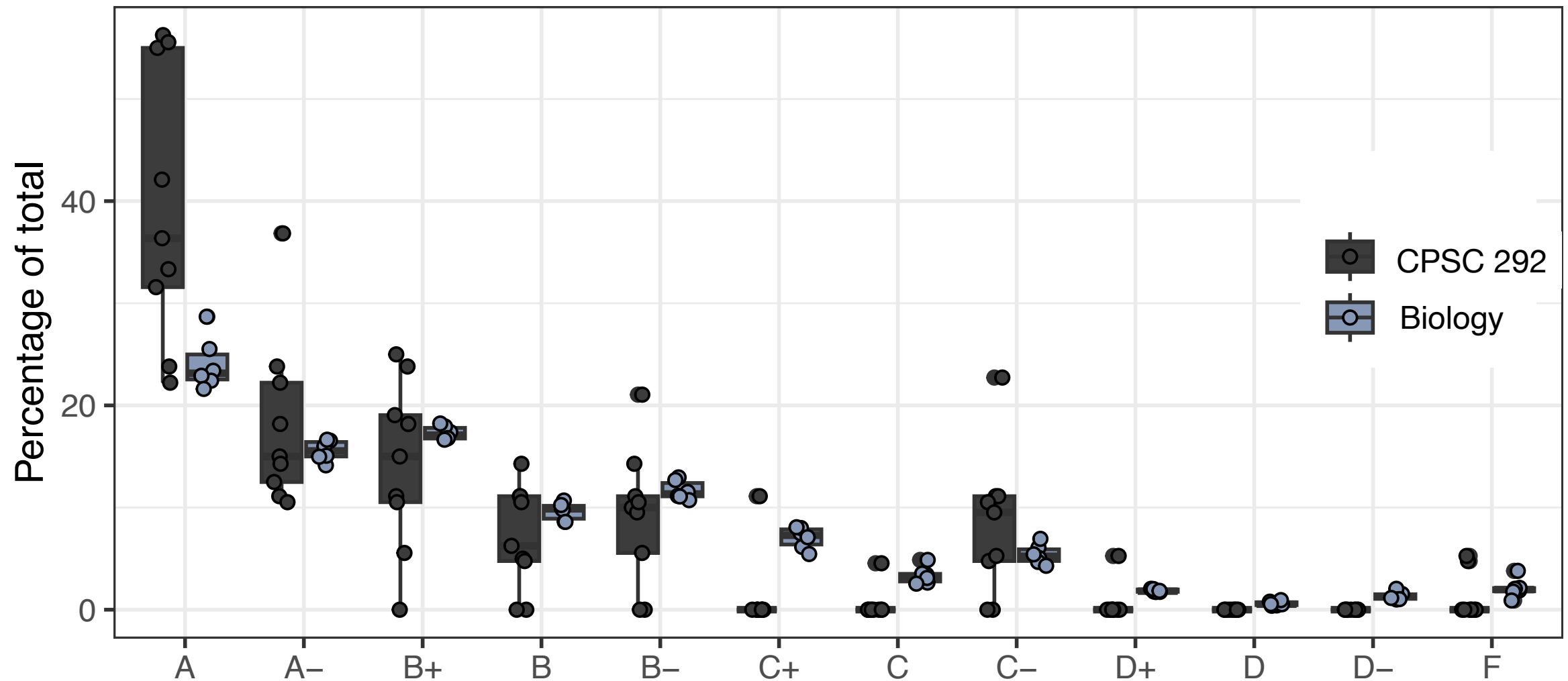
A Word about the Amount of Work...



- This course's effort is front-loaded.
- Unit 1 moves quickly, contains half of all course assignments!
- **Critically important to keep up with assignments in Unit 1.**

...and what that means about grades.

CPSC 292 versus Biological Sciences



More students earn A's in my course than a typical biology course!

Other Course Policies

- Masks are not required, but please be respectful of others' choices.
- Come prepared to learn and participate!
- Final project policy: you **MUST** participate in the final project.
- Group work is encouraged unless the assignment is an *individual evaluation*.
- Communication: Please Slack instead of email. Expect replies between 9 am and 5 pm during the regular work week.

Generative AI Use in this Course

- Gen AI is **limited use** in this course.
- **Gen AI Use is not allowed on any assignments.**

Example OK Uses

- brainstorming and refining ideas
- fine tuning visualizations or code
- debugging or explaining code

Example Prohibited Uses

- writing a draft of course work
- writing complete code blocks
- writing entire sentences, paragraphs, or papers

What if you need help?

- ***Please contact me!*** I want to help, whether it is situational, financial, or academic. I am prepared to be very flexible, including issuing course incompletes (which can be finished later).
- The Dean of Students can help connect you with services, no matter what type of problem you have!
- If you are struggling mentally, please talk to me or seek help through Student Psychological Counseling Services:
<https://www.chapman.edu/students/health-and-safety/psychological-counseling/>

Course Learning Objectives

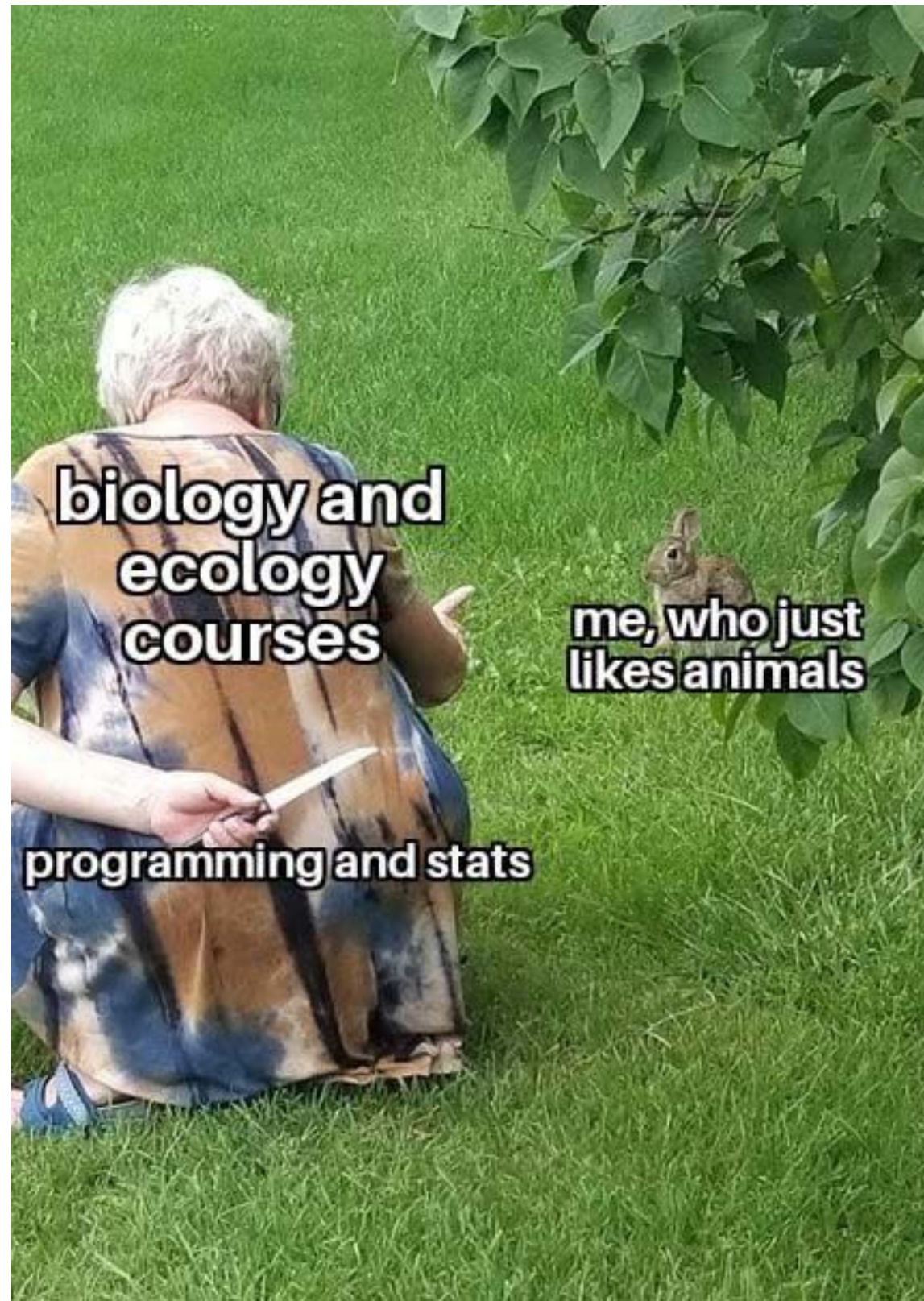


Main Learning Objectives:

1. Understand the basic structure and function of the *R* programming language.
2. Create visualizations and data analyses in the *R* programming language.
3. Independently perform basic data analysis and visualizations in a way that communicates ideas clearly.

Detailed learning objectives (and how they are assessed) are in the file CLO.pdf!

It's true. Sorry :(



Why Learn *R*?

- Biology today involves lots and lots and lots of data!
- Most disciplines require skill in handling and analyzing data.
- *R* is a high-level yet powerful programming language that can assist with statistics, analysis, and visualization.
- *R* is free and open-source, making analyses replicable.
- *R* is flexible and has a huge community working on new stuff!

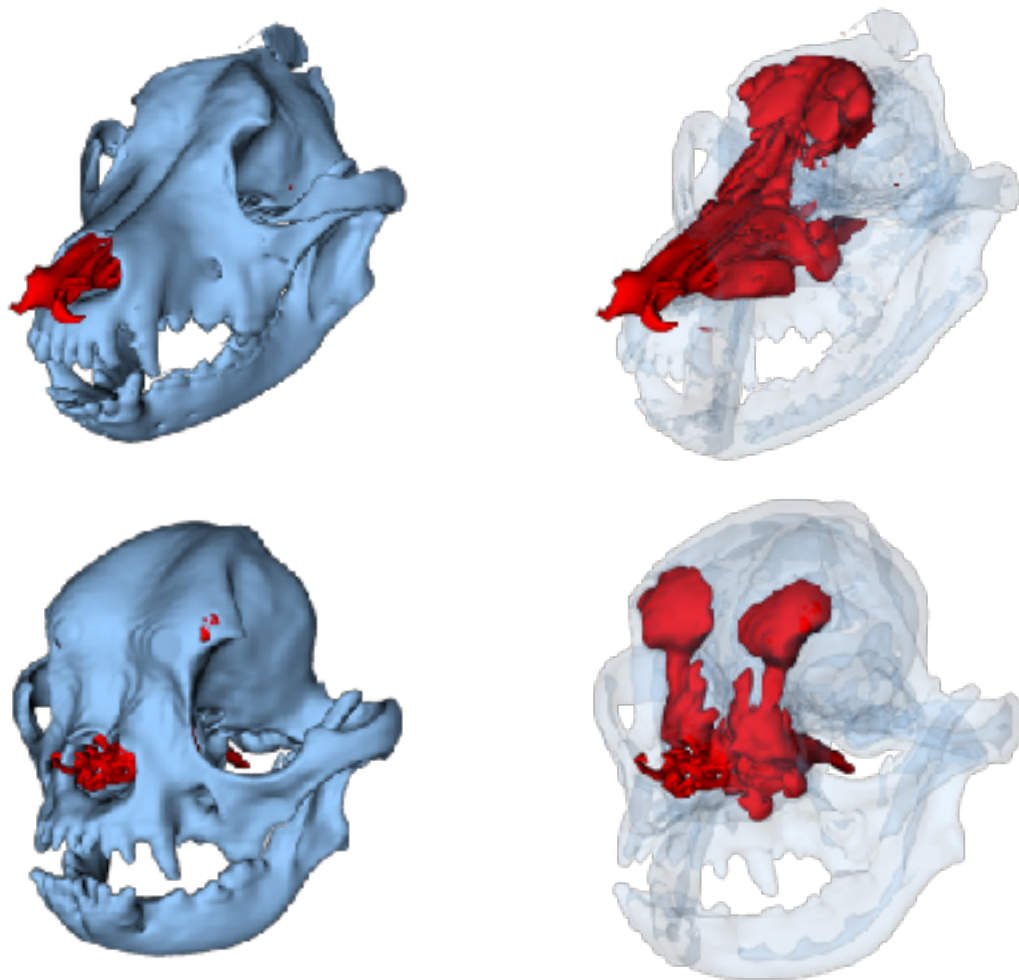


About me

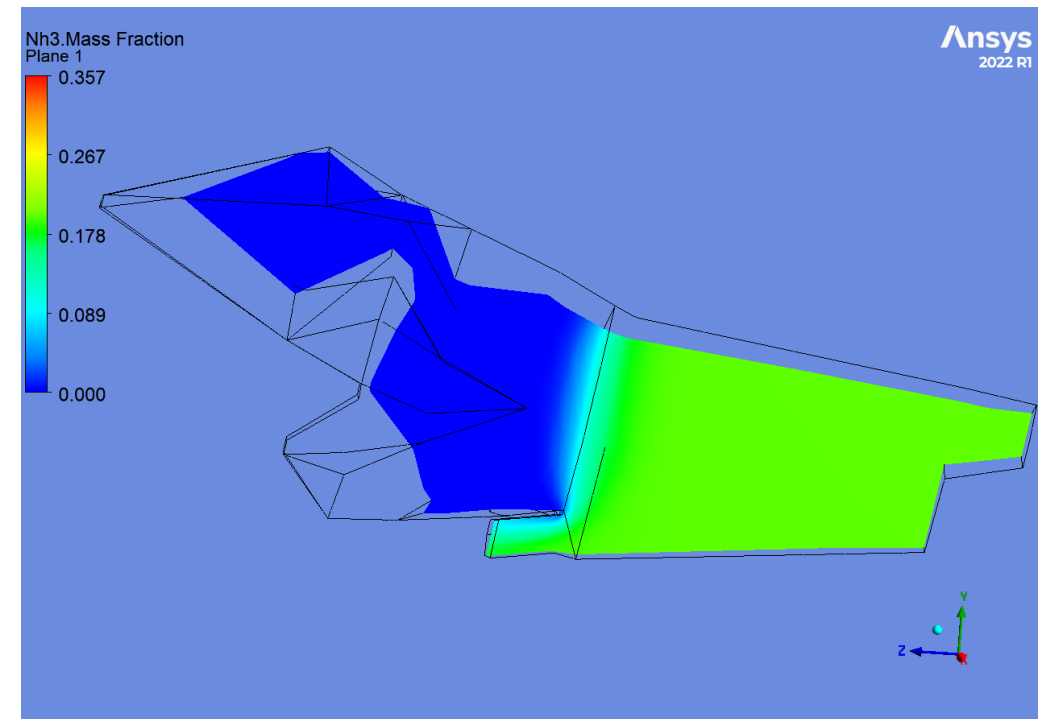
Lindsay Waldrop, Ph.D. (Biological Sciences) <http://waldroplab.com>

Exploring the evolution of biological fluid-structure interactions

Morphometrics



Computational fluid dynamics



Experimental fluid dynamics



How do you train a dog to find bombs (or whatever)?



Frame 1140, Time 00:00:38.038

How do you train a dog to find bombs (or whatever)?



How do you train a dog to find bombs (or whatever)?

You split the behavior into little, tiny pieces and teach them a bit at a time!

1. Imprinting

**“This smell means: FREEZE
and then fun!”**

2. Discrimination

“This smell not that one.”

3. Generalizing

**“Gotta keep getting closer to The
Smell no matter what.”**

Imprinting

This step teaches the dog that the game means if you smell this special odor, you FREEZE and then get a treat or toy.

IT MUST BE EASY AND FUN!



Takeaways:

Short sessions

Easy, quick success

Lots of feedback

**Individuals work at
different paces**

Discrimination

This step teaches the dog that the game means the target smell, not any other smell.

Errors are REQUIRED for this step!



Takeaways:

Make things harder a little bit at a time

Try stuff - it doesn't matter if it's wrong

Individuals work at different paces



Generalization

This step teaches the dog that the game is the same regardless of the environment.

It takes the longest and is the most difficult!



Takeaways:

Try skills in different situations to learn

It's ok to struggle and look for support

It's ok to get a little overwhelmed, just keep at it

It's ok to end a little early

Progress, not perfection



Don't skip the simple stuff!

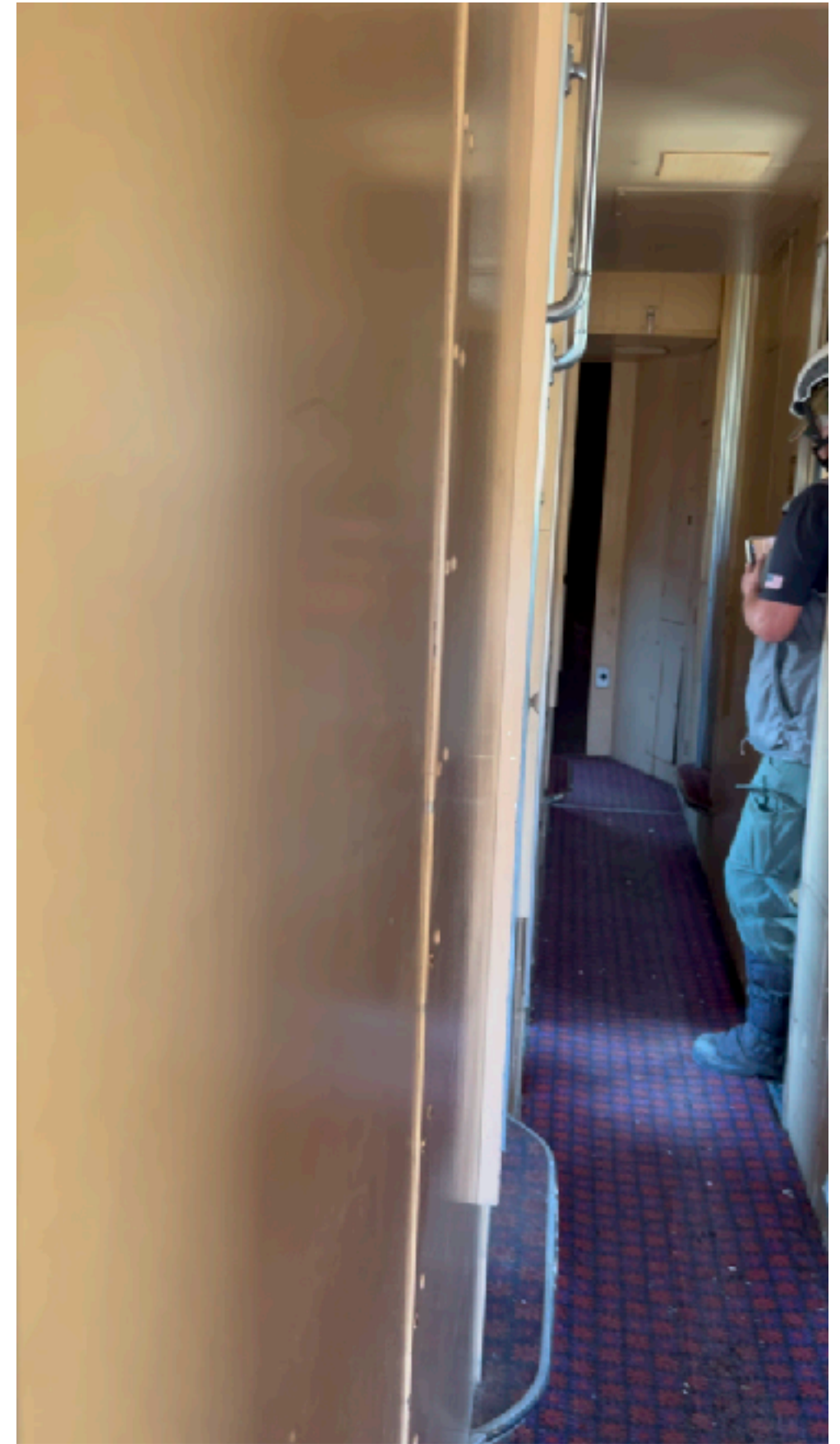
Take it at your own pace!

It's ok to struggle and look for support!

Where we're at today



The Real Pros (Search Dog Foundation)



Downloading *R* and RStudio



Download *R*:
<https://www.r-project.org/>

1. Go to <https://cloud.r-project.org/>
2. Select your operating system.
3. Select the latest release that is “notarized and signed.”
4. Save and open the file, follow the instructions to install.

NOTE: be sure to select the correct chip type for Macs!



Download RStudio:
<https://rstudio.com/products/rstudio/download/>

1. Select the RStudio Desktop version.
2. Download, open, and follow instructions to install.
3. Open RStudio to get started!

Action Items

- 1. Have R and RStudio installed on your personal computer by Wed 9/3!**
- 2. If you are having trouble with a laptop, please let me know!**